CIUK 2026, UCL challenge

Which patients' E. coli infections resist co-amoxiclav?

Tree models, run on UCL's Condenser cloud

UCLuster Too

Ada Ge · Bobby Ta · Eric Huang · James Loke

Q1. What is cloud compute?

Cloud computing: renting computers over the internet when you need them

Our code	**we run this**
Our virtual machine: a computer made in software	Condenser gave us this
Hardware shared with other users	**UCL runs this**

Condenser is UCL's private cloud for research computing. This split is called infrastructure as a service.

Source: condenser.arc.ucl.ac.uk/about

Q2. What are key data security considerations for cloud computing?

Security on Condenser is split between UCL and us

UCL

- Hardware in UCL's own facilities
- Each team's machines kept apart from other teams'

Us

- Log in only through UCL's JupyterHub, with an SSH key, over encrypted connections
- No passwords or keys in our public repository

> Condenser's own page: "not certified for the secure storage and processing of sensitive data"

Source: condenser.arc.ucl.ac.uk/about, checked 29 Oct 2026

Q3. The data file was world-readable. Why is this not desirable? Did we change it, how and why?

Any user on our machine could read the data file, so we changed its permissions

```
# before: any user on the machine can read the file
-rw-r--r--. 1 cc cc 94198 Oct 26 16:44 dig_health_hub_amr.csv

# moved it into data/, then
$ chmod 700 data
$ chmod 600 data/dig_health_hub_amr.csv

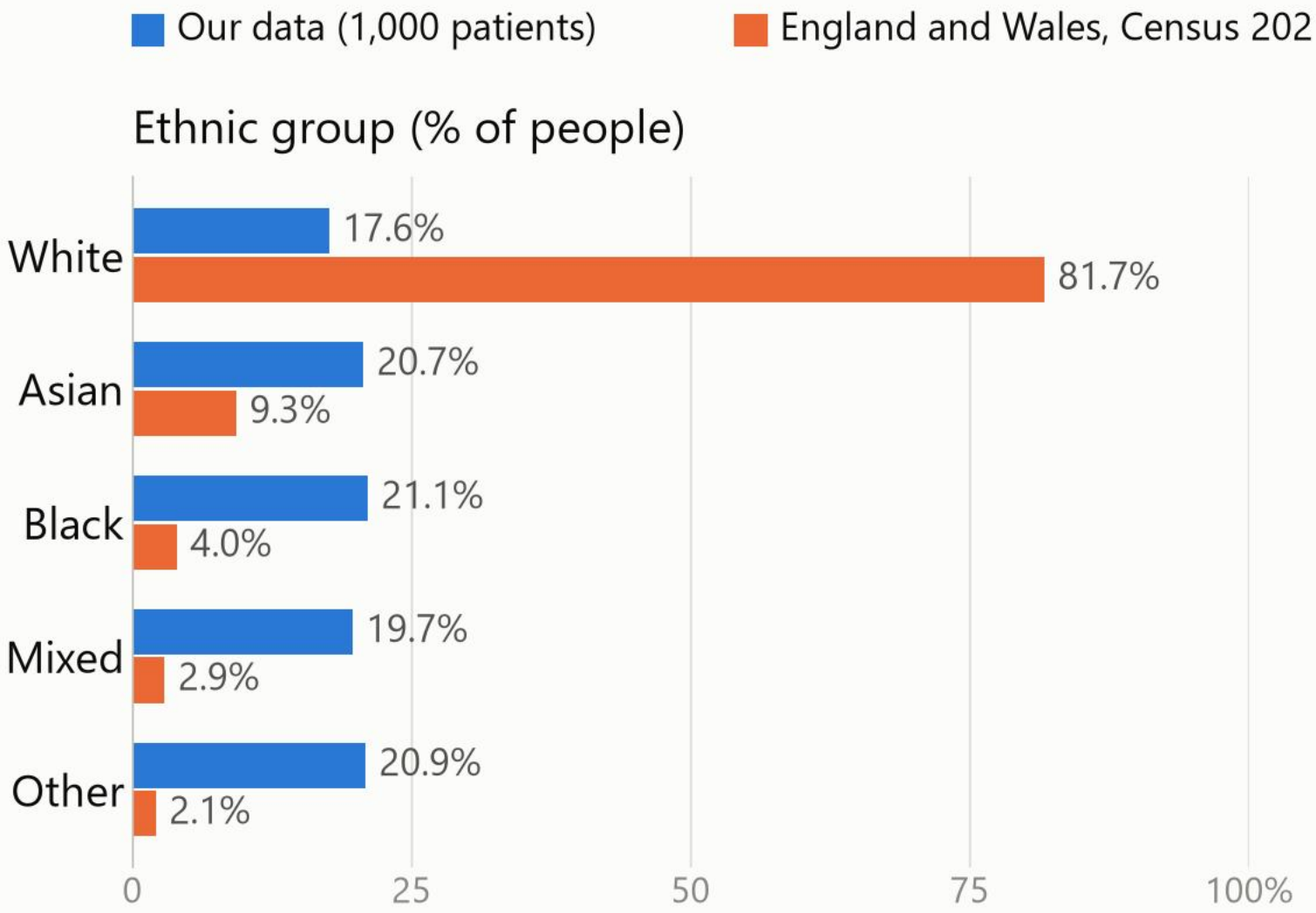
# after: only our account (cc) can open the folder or read the file
drwx------. 2 cc cc    36 Oct 27 09:51 data
-rw-------. 1 cc cc 94198 Oct 26 16:44 dig_health_hub_amr.csv
```

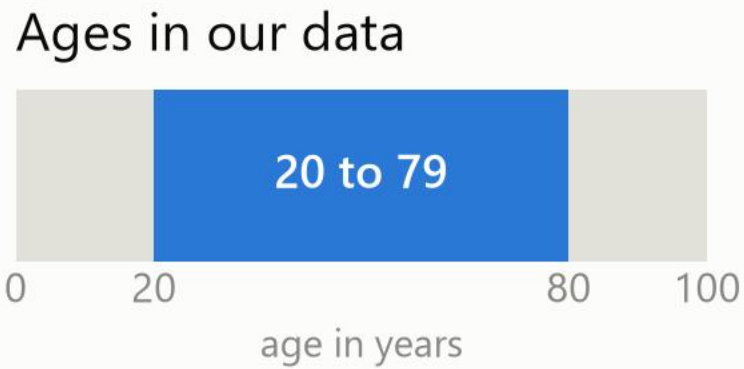
Our scripts also set `umask 027`, so the files they write are closed to other users.

Retyped from notes/permissions-before.txt and permissions-after.txt (our VM, 27 Oct 2026)

Is our data representative?

The 1,000 synthetic patients, compared with the population of England and Wales





Nobody under 20 or over 79, so the model knows nothing about them

Sources: our data from dig_health_hub_amr.csv (team notes, 25 Oct 2026); population from ONS Census 2021, England and Wales (5 ethnic groups).

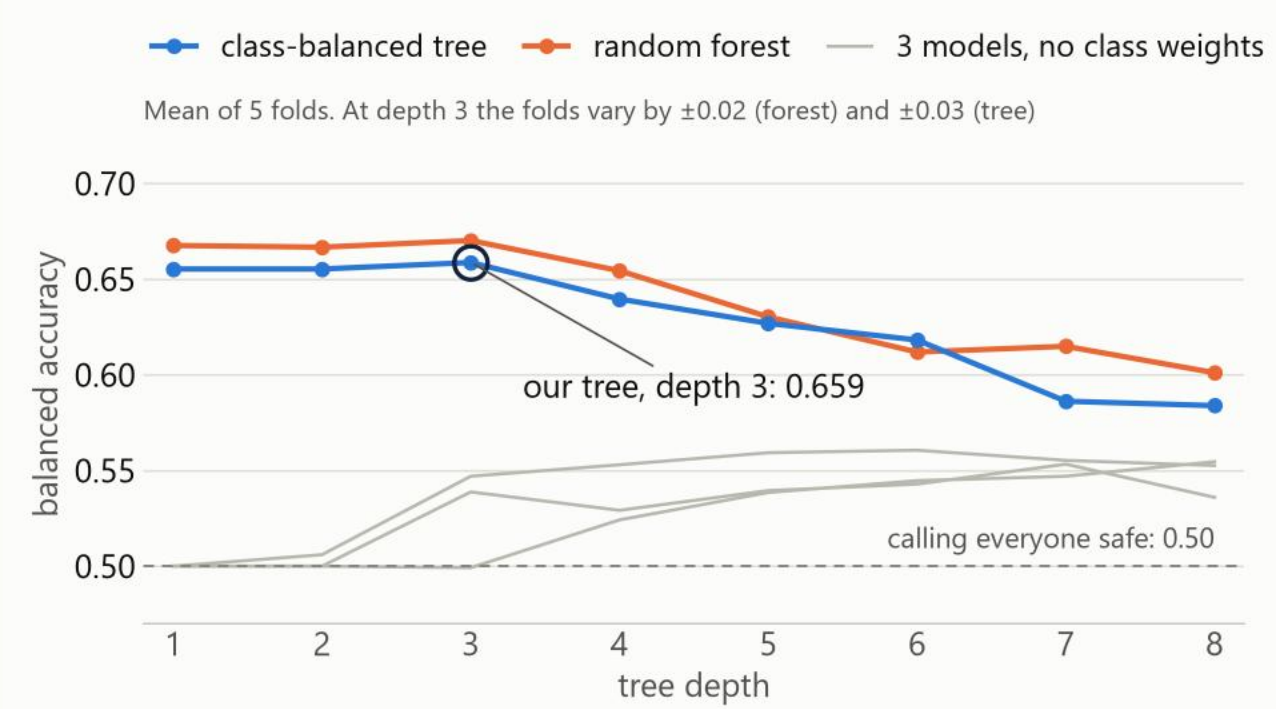
Q5. Why would we want to use synthetic data instead of actual patient data?

Synthetic data is made to mimic real data, with no real people in it

Real records	Protected by UK GDPR. Patients may not have agreed to this use.
Synthetic data	Far less privacy risk, even on a shared cloud machine.

Q6. What is our final tree model, and why is it the most optimal?

Calling everyone safe scores 76% accuracy, so we ranked models on balanced accuracy instead

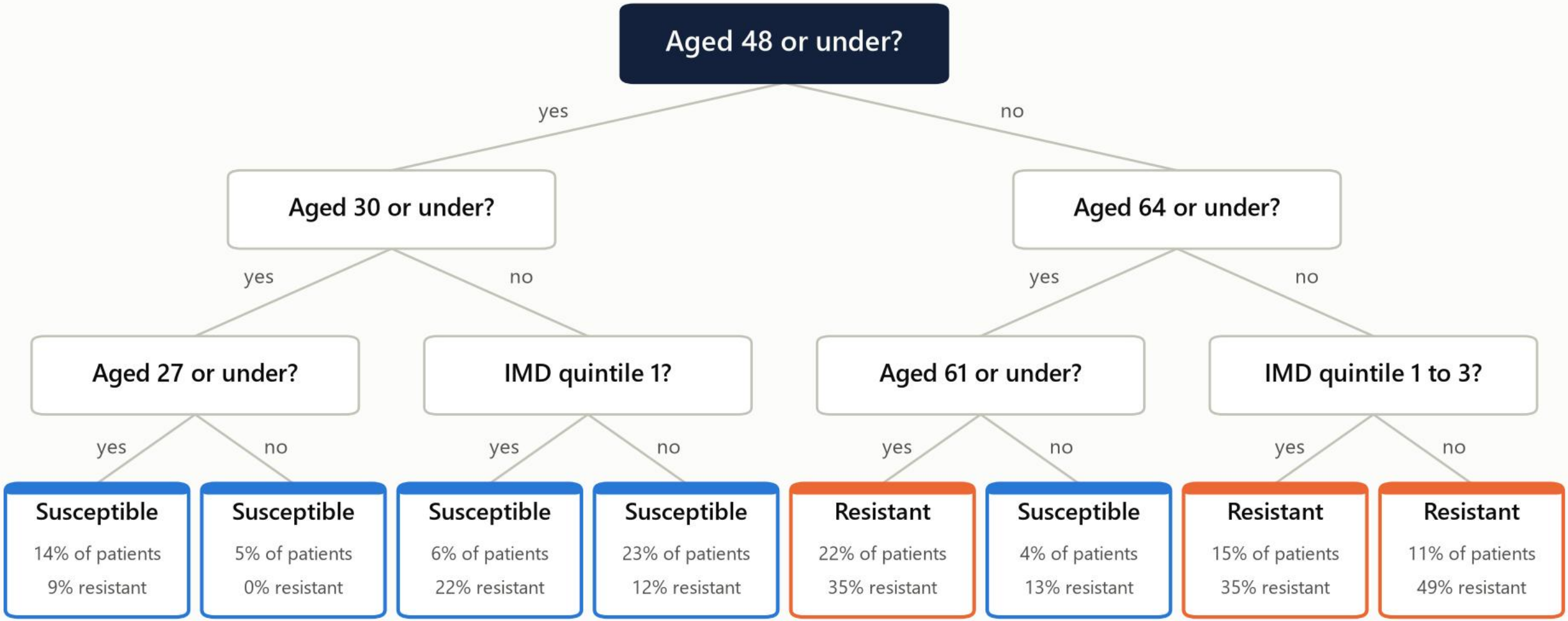


Balanced accuracy	5-fold CV	Test
Calling everyone safe	0.50	0.50
Our tree, depth 3	**0.659**	**0.622**
Random forest, 200 trees	0.670	0.675

CV: 5 folds of the 750 training patients. Test: 250 patients held back until the end. outputs/model_comparison.csv

Our final model: a class-balanced decision tree, depth 3

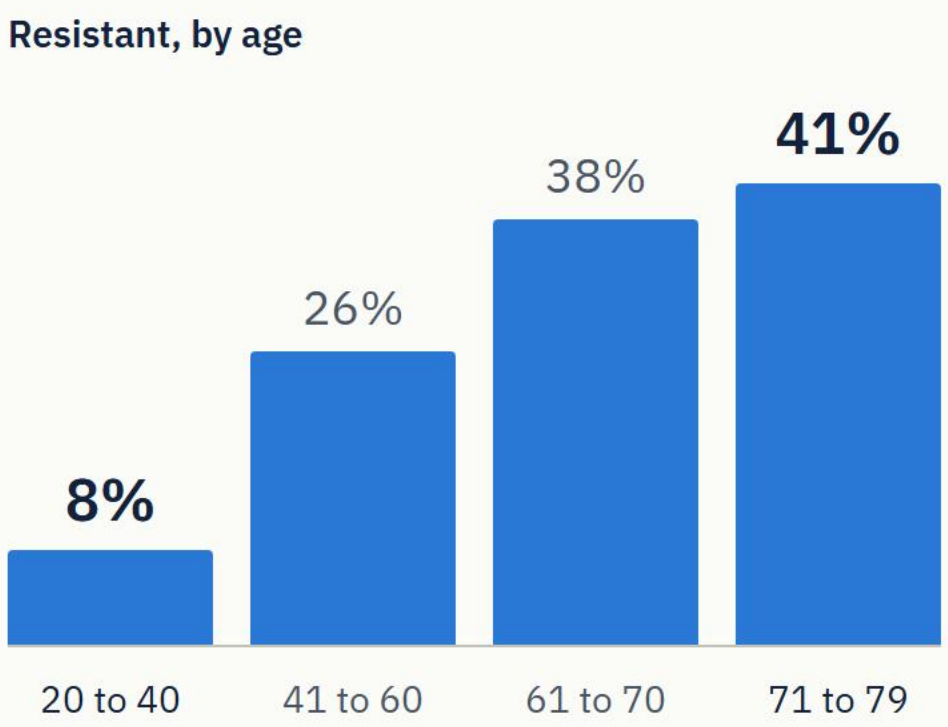
Each end box: what the tree predicts, its share of the 750 training patients, and how many of them were resistant.
A group is flagged resistant when its rate is above the overall 24%. IMD: deprivation quintile, 1 = most deprived.

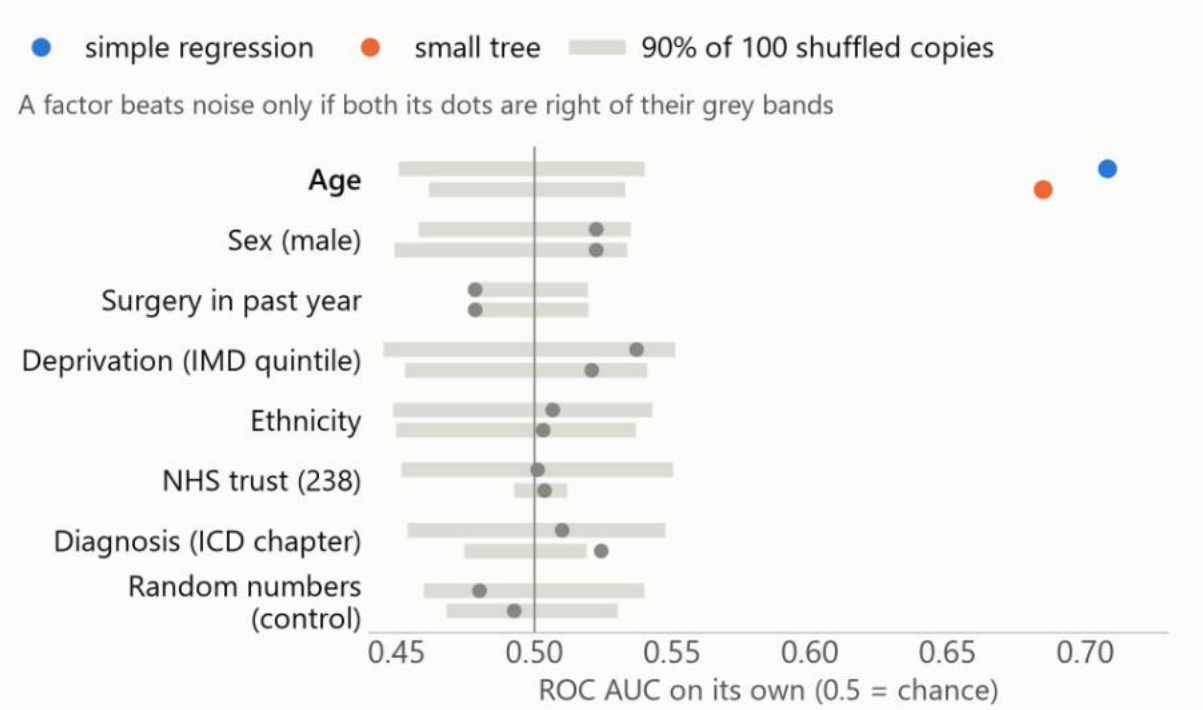


Read as a rule: over 48, flagged resistant, except ages 62 to 64. Aged 48 or under, susceptible.

Q6. What is our final tree model, and why is it the most optimal?

Resistance rises with age, and only age beats noise in both tests



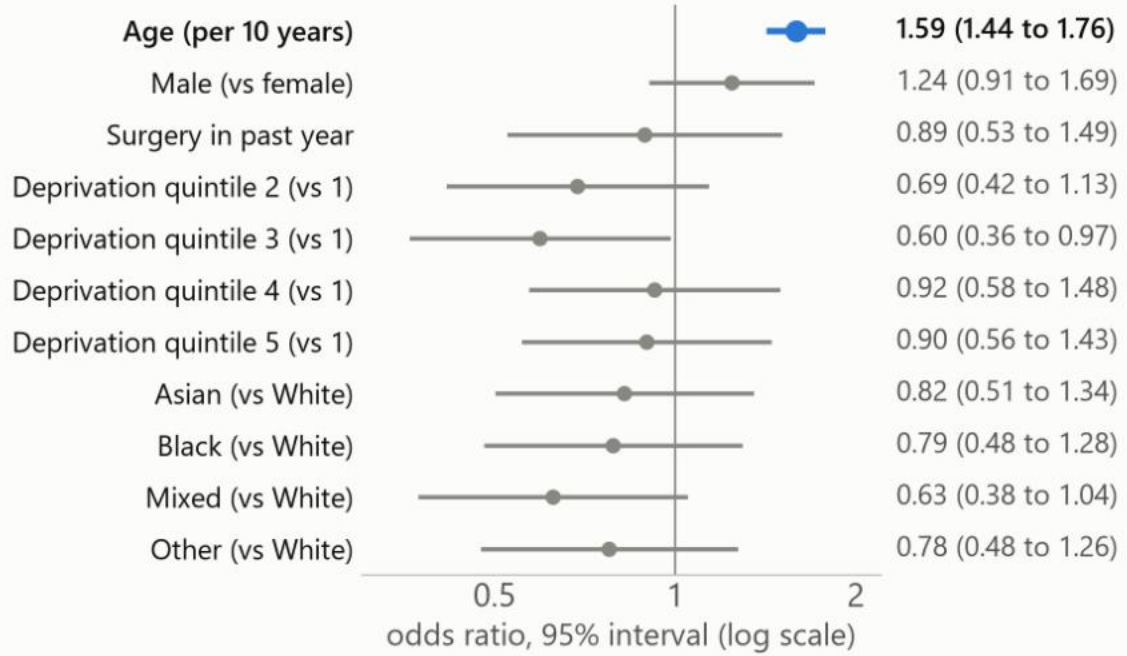


1,000 synthetic patients (notes/2026-10-29-model-checks.md). Right: src/risk_factors.py, each factor alone vs 100 shuffled copies of it

Q7. What other modelling techniques could we use, and why?

Logistic regression matched our tree

Separate fit on all 1,000 patients, five factors together. Quintile 1 = most deprived.



Balanced accuracy on age and deprivation, same 5 folds as our tree

Logistic regression	**0.660**
Our tree	**0.659**

Fits risk that rises smoothly with age, and gives odds ratios. A neural network isn't needed for 1,000 patients and one clear risk factor.

Deprivation quintile 3 just clears 1, but quintiles 2 to 5 show no trend.

Scores: age and deprivation (notes/2026-10-29-model-checks.md). Odds ratios: outputs/risk_factors_adjusted_odds_ratios.csv

Q8. How did we work as a team?

Who did what

Bobby	Cloud machine, environment and ReadMe
James	Model script
Ada	Explored the data and chose how to evaluate the models
Eric	Led the video

We paired up to cover each other and checked in for 15 minutes each evening. The plan, role guides and work-session log are in our GitHub repository.

Q8. How did we work as a team?

Problems we ran into

Problem	What we did
Login details reached only one of us, with no username	Shared that login as instructed, and found the username in the organisers' access guide
No git on the cloud machine	Edited on laptops, pushed to GitHub, fetched files with curl
Our first model flagged nobody as resistant	Switched to balanced accuracy and class weights

Age is the only clear risk factor in our synthetic data

Final model: a class-balanced decision tree, depth 3. Balanced accuracy: 0.66 in cross-validation, 0.62 on held-out patients.

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`github.com/adellama75/cluster-challenge-ucl`